

SK016/2
Chemistry
Paper 2
Semester I
Session 2014/2015
2½ hours

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Kimia
Kertas 2
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2½ jam



KEMENTERIAN
PENDIDIKAN
MALAYSIA

BAHAGIAN MATRIKULASI
MATRICULATION DIVISION

PEPERIKSAAN SEMESTER PROGRAM MATRIKULASI
MATRICULATION PROGRAMME EXAMINATION

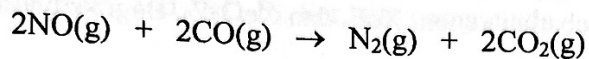
KIMIA
Kertas 2
2½ jam

JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU.
DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.

SECTION A [60 marks]

Answer *all* questions in this section.

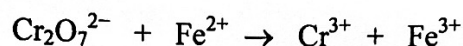
- 1 (a) Consider the following reaction:



If 300.00 g of CO has reacted, calculate the mass percentage, % w/w, of CO_2 in the mixture of product.

[5 marks]

- (b) A 30.00 mL of 0.025 M sodium dichromate, $\text{Na}_2\text{Cr}_2\text{O}_7$, solution is titrated with iron(II) sulphate, Fe_2SO_4 solution in acidic condition, according to the following reaction:



The titration requires 40.00 mL of Fe_2SO_4 solution to reach the end point.

- (i) Balance the redox equation.

[3 marks]

- (ii) Calculate the mass of $\text{Na}_2\text{Cr}_2\text{O}_7$ needed to prepare a 0.025 M solution in a 50 mL volumetric flask.

[3 marks]

- (iii) Determine the molarity of the Fe^{2+} solution.

[4 marks]

- 2 (a) The formula of formic acid is HCO_2H . One of the carbon-oxygen bond lengths in this molecule is $1.36 \text{ \AA}$ while the other is $1.23 \text{ \AA}$. Draw the Lewis structure of this molecule and label these bonds. [5 marks]
- (b) Xenon can be covalently bonded to fluorine and oxygen to form xenon compounds, XeF_4 and XeO_2F_2 . For both compounds,
- (i) determine the number of bonding electron pair(s) and lone electron pair(s) around the central atom xenon. [4 marks]
 - (ii) state the molecular geometry. [2 marks]
 - (iii) determine the hybridisation of xenon atom. [2 marks]
 - (iv) predict their polarity. [2 marks]
- 3 (a) Limestone, CaCO_3 , decomposed to solid calcium oxide and carbon dioxide gas when heated at high temperature. At 30°C , a volume of 107.3 mL of the gas was collected by displacement of water with a total pressure of 1 atm . Calculate;
- (i) the number of moles of carbon dioxide produced. [5 marks]
 - (ii) the mass of limestone decomposed. [4 marks]
- [Vapor pressure of water at 30°C is 31.8 mmHg]
- (b) Briefly describe three types of crystalline solids in terms of their interparticle forces. [6 marks]

- 4 (a) HA is a weak acid with an acid dissociation constant, $K_a = 2.95 \times 10^{-8}$. If the concentration of the acid is 1.12 M, calculate
- (i) pH of the solution. [6 marks]
 - (ii) percent dissociation. [2 marks]
- (b) The solubility product constant, K_{sp} , of calcium phosphate, $\text{Ca}_3(\text{PO}_4)_2$, in pure water is 1.2×10^{-26} at 25°C .
- (i) Write the expression for the solubility product constant. [2 marks]
 - (ii) Calculate the molar solubility of calcium phosphate and concentration of each ion. [5 marks]

SECTION B [40 marks]

Answer only two questions in this section.

- 5 The ion of atom *X* has 8 outermost electrons and 10 inner electrons with a charge of -1. Discuss all the rule(s) and principle(s) used to fill the electrons in the orbital of atom *X*. Explain the change in the radius of atom *X* as it changes from a neutral atom to a negatively charged ion.

Atoms *X*, *Y* and *Z* are in periods n , $n+1$ and $n+2$, respectively. These atoms are also in the same group. Discuss the trend in electronegativity exhibited by these atoms.

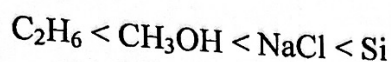
[20 marks]

- 6 Draw and explain the structures of BH_3 , NH_3 and PH_3 using valence shell electron pair repulsion theory. Also, show the overlapping of orbitals in the PH_3 molecule.

The boiling points of BH_3 , NH_3 and PH_3 are 173 K, 240 K and 185 K, respectively. Explain why the boiling point of PH_3 is greater than BH_3 but lower than NH_3 .

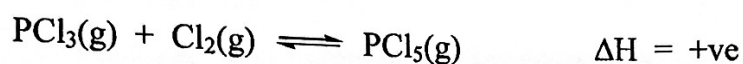
[20 marks]

- 7 (a) The melting points of C_2H_6 , CH_3OH , $NaCl$ and Si increase in the order as shown below. Explain the trend.



[5 marks]

- (b) A 10.00-L vessel contains 0.0681 mol phosphorous trichloride, PCl_3 , and 0.2056 mol chlorine, Cl_2 , at $250^\circ C$. If 0.0316 mol phosphorous pentachloride, PCl_5 , is produced at equilibrium, calculate K_p for this reaction.



Discuss **four** factors that can increase the amount of PCl_5 produced.

[15 marks]

- 8 A sample of 0.1000 g of $NaOH$ is dissolved in 25 mL of water and titrated stepwise until 30 mL of 0.100 M HCl is added. Using these data, describe the titration process from beginning till the end. Sketch a graph showing the titration curve for this process.

If HCl is replaced by a weak monoprotic acid, HY , sketch the expected titration curve on the same graph as above.

[20 marks]

END OF QUESTION PAPER